Test Dimensionality

- Imagine that we wish to use a personality inventory that includes the following six adjectives:
  1. Talkative
  2. Assertive
  3. Imaginative
  4. Creative
  5. Outgoing
  6. Intellectual
- Participants are asked to consider the degree to which each adjective describes their personality in general

- What exactly does this inventory measure?
- Does it measure six separate facets of personality, with each facet being reflected by a single adjective?
- Does it measure a single construct?
- If so, then what is that construct – what do these six adjectives have in common as a psychological characteristic or dimension?
- Are there two or three separate dimensions reflected within these six adjectives?
- How will this questionnaire be scored?

- Group them into clusters that seem to share some common meaning / in terms of their similarity to each other
- For example,
  - Talkative
  - Assertive
  - Outgoing

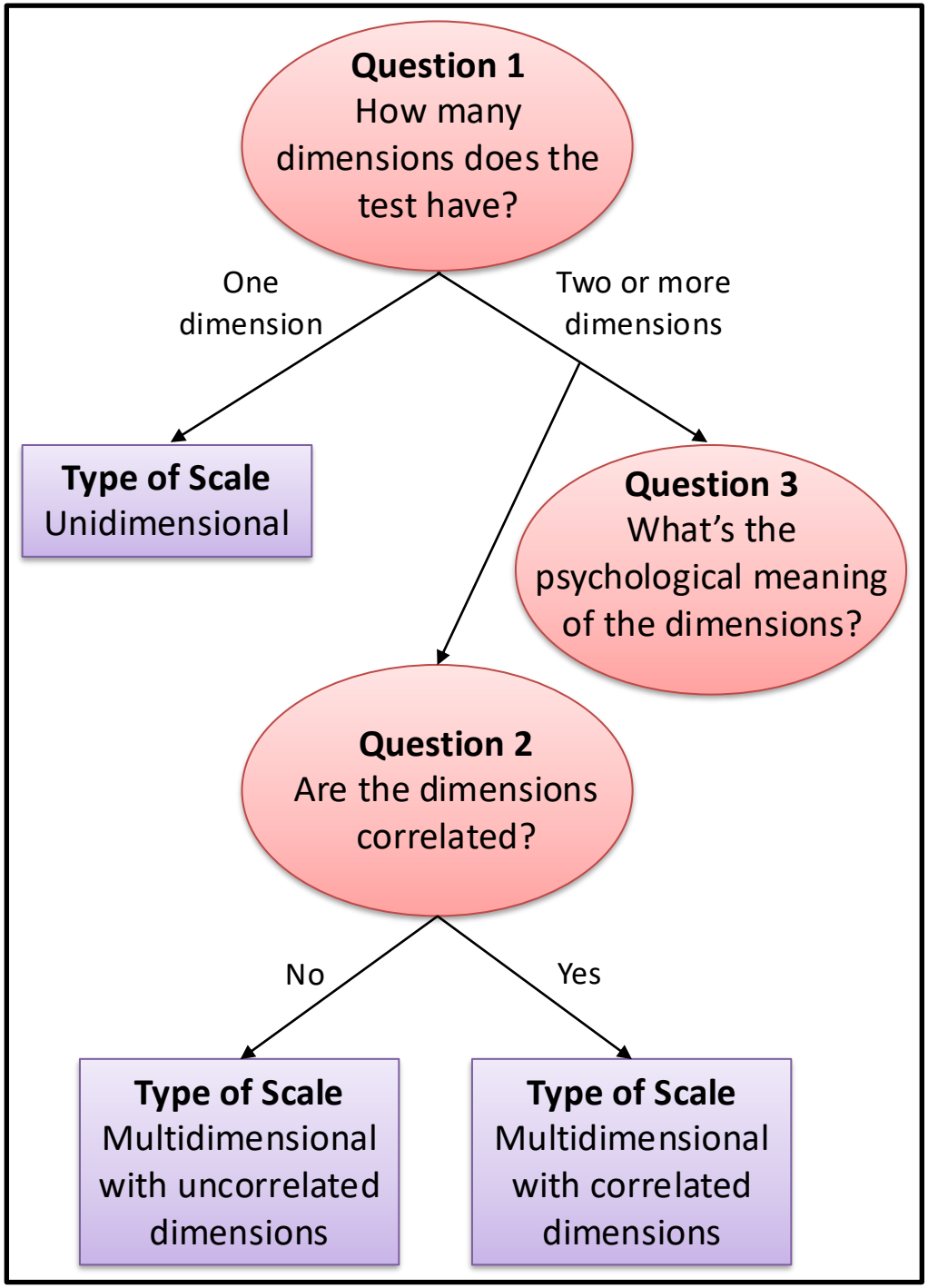
  → three variations on one attribute "**extraversion**"

  - Imaginative
  - Creative
  - Intellectual

  → three variations on another attribute "**openness to experience**"

- From this perspective, responses to these six personality adjectives reflect two basic dimensions:
  - One set of responses that are a function of an extraversion dimension
  - One set of responses that are the result of an openness to experience dimension
- This example illustrates the issue of test dimensionality, which is a fundamental consideration in test development, evaluation, and use

- There are at least three fundamental questions regarding the dimensionality of a test
- The answers to these questions have important implications for
  - evaluating the psychometric properties of any behavioral test
  - appropriately scoring on a test
  - the proper interpretation of test scores
- The goal is to discuss
  - the concept of dimensionality
  - the key questions related to dimensionality
  - the implications that dimensionality has for test construction, evaluation, use, and interpretation



Three fundamental questions:

- How many dimensions does the test have?
- Are the dimensions correlated?
- What's the psychological meaning of the dimensions?

As shown in figure, the answers to the three key questions lead to three main types of test:

- Unidimensional tests
- Multidimensional tests with correlated dimensions
- Multidimensional test with uncorrelated dimensions

Dimensionality

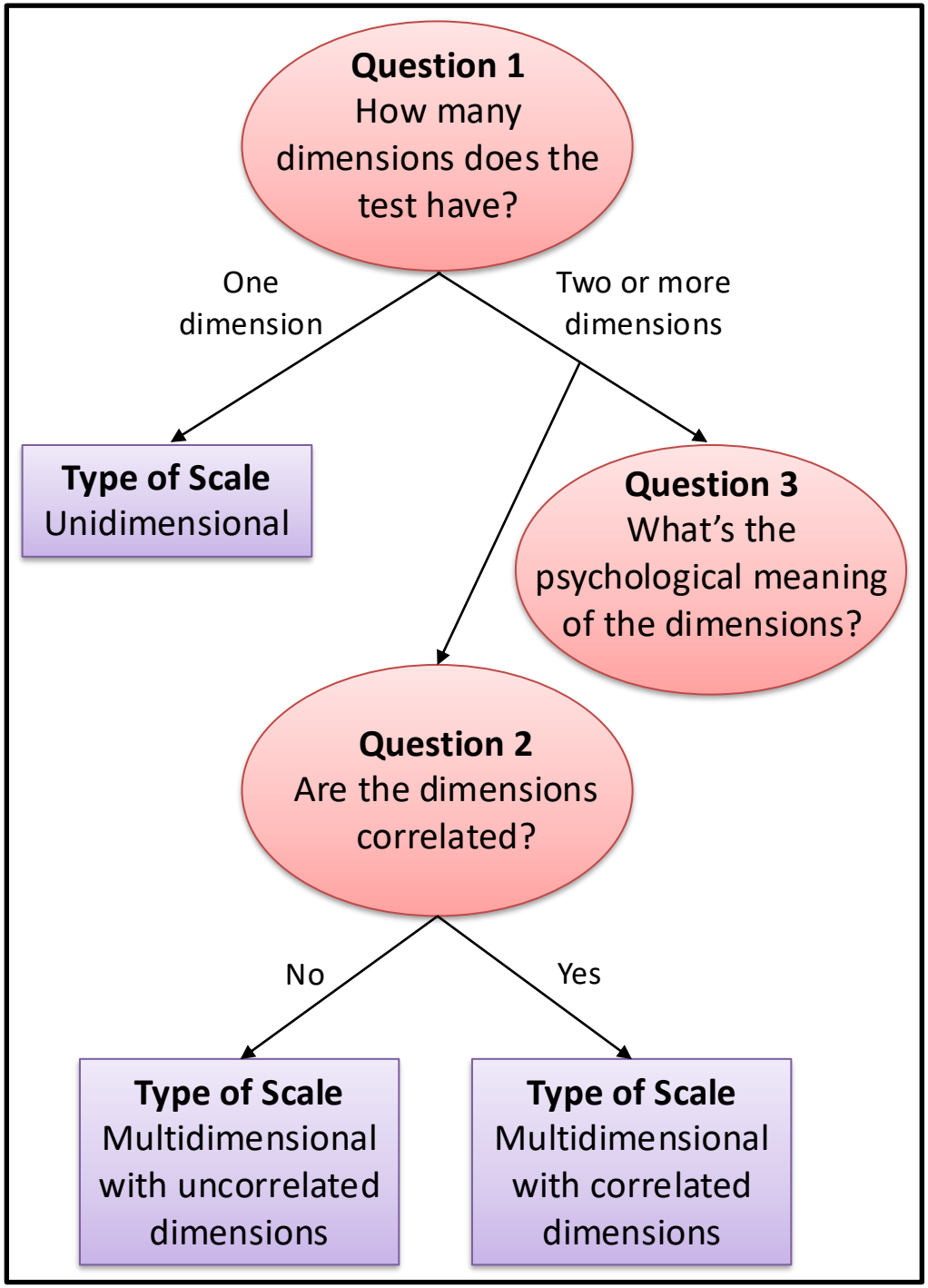
- If you step on a bathroom scale, the resulting number on the scale is a value that tells you something about one of your physical attributes or features – your weight
- As a human being, you have many other physical attributes, including height, skin color, length of hair, and so on
- When you weigh yourself, the number that represents your weight should not be influenced by attributes such as your hair color, your height, or your age
- The “score” on the bathroom scale should reflect one and only one physical dimension

- Similarly, if we have a psychological test that yields some kind of number, then we would like to think of the number as a value representing a single psychological feature or attribute
- For example, suppose you had a test of courage
- If you have a test that produces scores that can be treated as if they are real numbers, then a person's score on the test might indicate the amount of courage that he or she had
- We could then think of courage as an attribute of that person and the test score as an indication of the amount of the person's courage
- The score on the courage test should reflect one and only one psychological dimension

- As a general rule, when we measure a physical or psychological attribute of an object or a person, we intend to measure a single attribute of that object or person
- In the case of weight, we try to measure weight so that our measurement is not affected by other attributes of the person being measured
- If we measure someone's weight, measure his or her hair length, and then add those scores together to form a "total" score, does it make sense?
- Clearly, the total score would be a blend of two physical dimensions that are almost totally unrelated to each other, and the combination of the two scores would have no clear interpretation
- That is, the total score would not have clear reference to a single physical attribute and thus would have no clear meaning

- Similarly, it would not be reasonable to measure someone's courage, measure his or her verbal skill, and then add those two scores together to form a "total" score
- Again, the total score would be blend of two dimensions that are clearly unrelated to each other
- Combining test scores from two independent psychological attributes produce a total score that has no clear meaning

- The scores from a wide variety of psychological tests are based on multiple questions or test items
  - For example, personality tests range in length from 5 or fewer questions to several hundred questions
- In scoring such tests, item responses are combined in some way, usually by computing one or more scores of some kind, and these combined scores are used to reflect the psychological attribute(s) of interest
- These scores are referred to as ***composite scores***, and ideally, a composite score reflects one and only one dimension
- However, a test may include items that reflect more than a single dimension



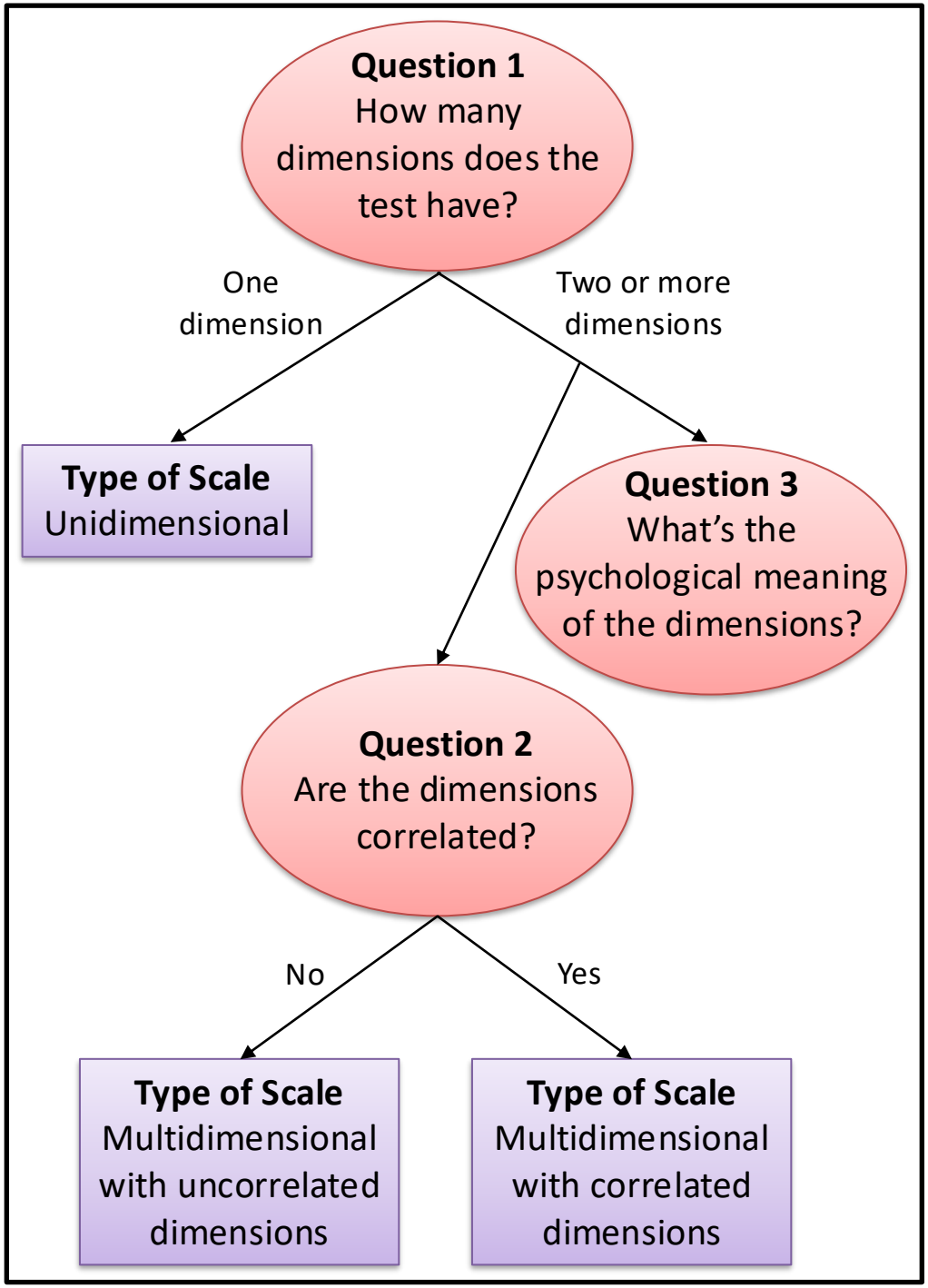
Three Dimensionality Questions

How many dimensions does the test have?

Some tests reflect one and only one dimension, while others reflect two or more psychological dimensions

This issue is important because;

Each dimension of a test is likely to be scored separately, with each dimension requiring its own psychometric analysis



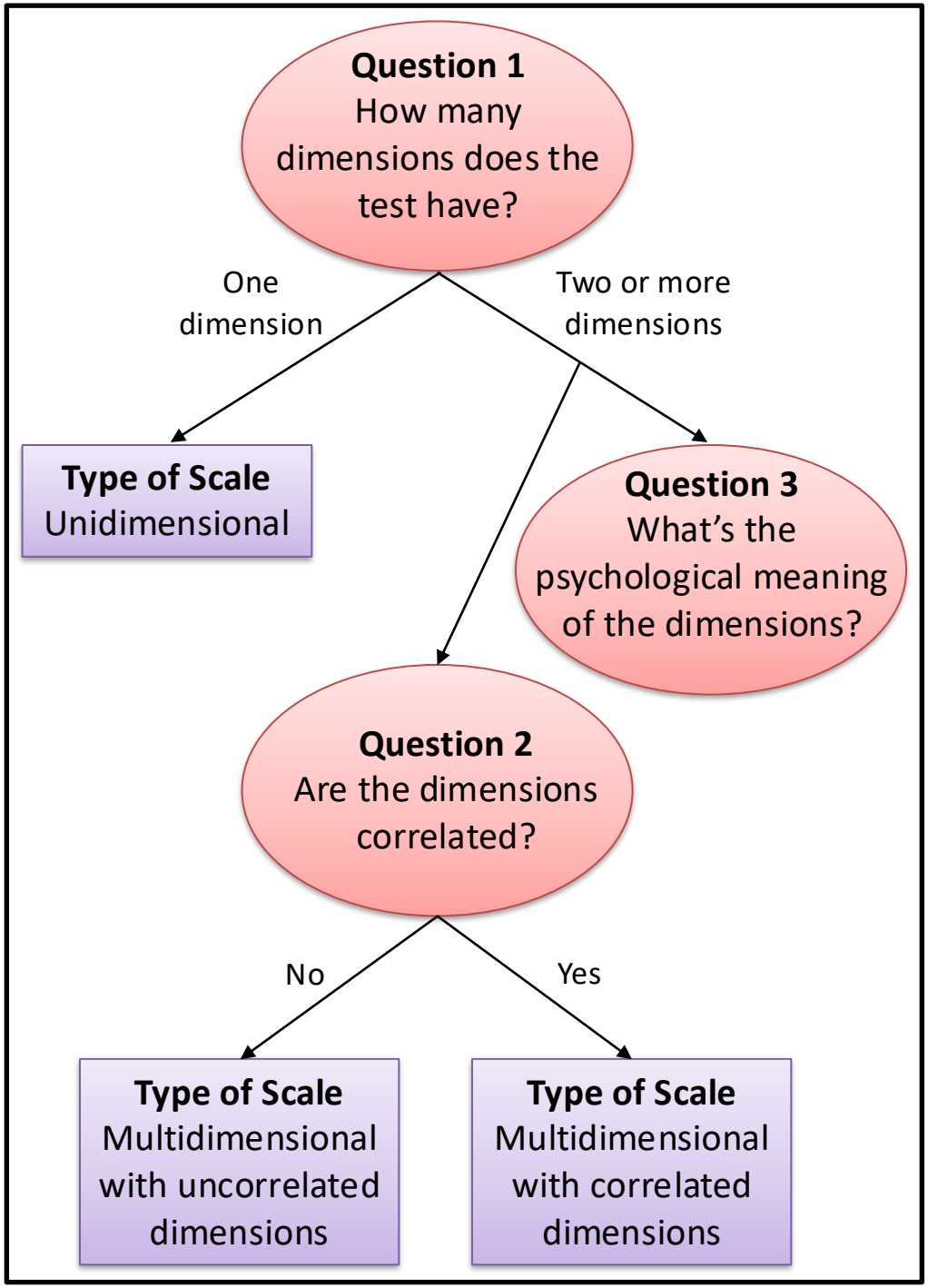
Three Dimensionality Questions

If a test more than one dimension, then are those dimensions correlated with each other?

Some tests have sereval dimensions that are somewhat related to each other while other tests have several dimensions that are essentially independent

This issue is important because;

The nature of the associations among a test's dimensions has implications for the meaningfulness of a "total score" for a test



Three Dimensionality Questions

If a test more than one dimension, then what are those dimensions?

For example, in the personality test example, does the first dimension reflect the psychological attribute of extraversion or some other attribute?

This issue is important because;

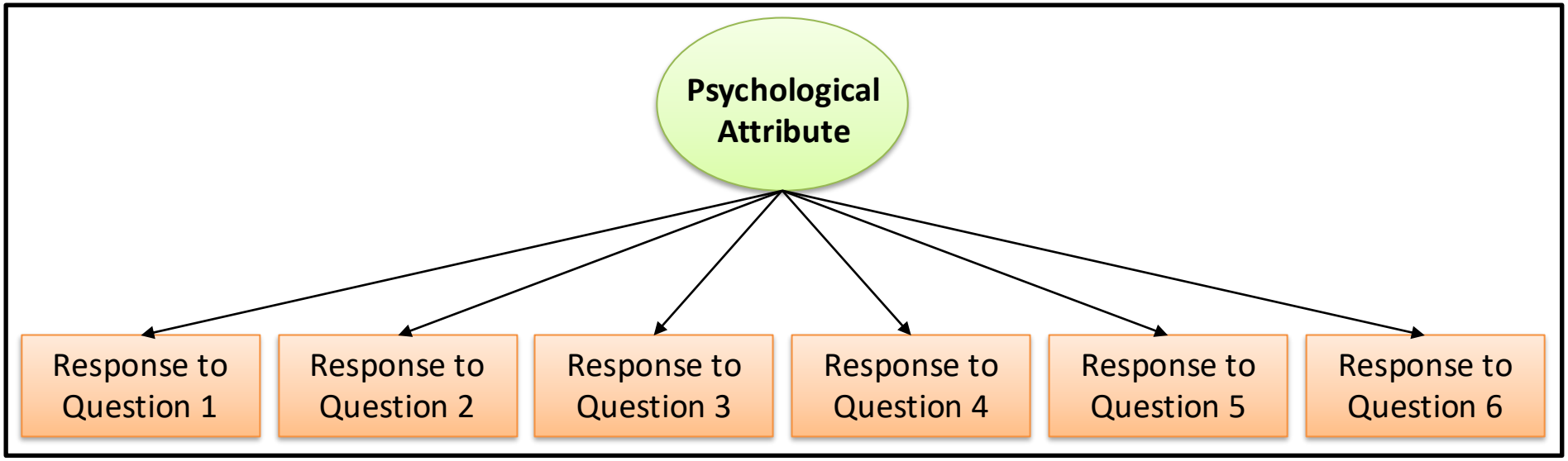
If we score and interpret a dimension of a test, we must understand the score's psychological meaning

Unidimensional Tests

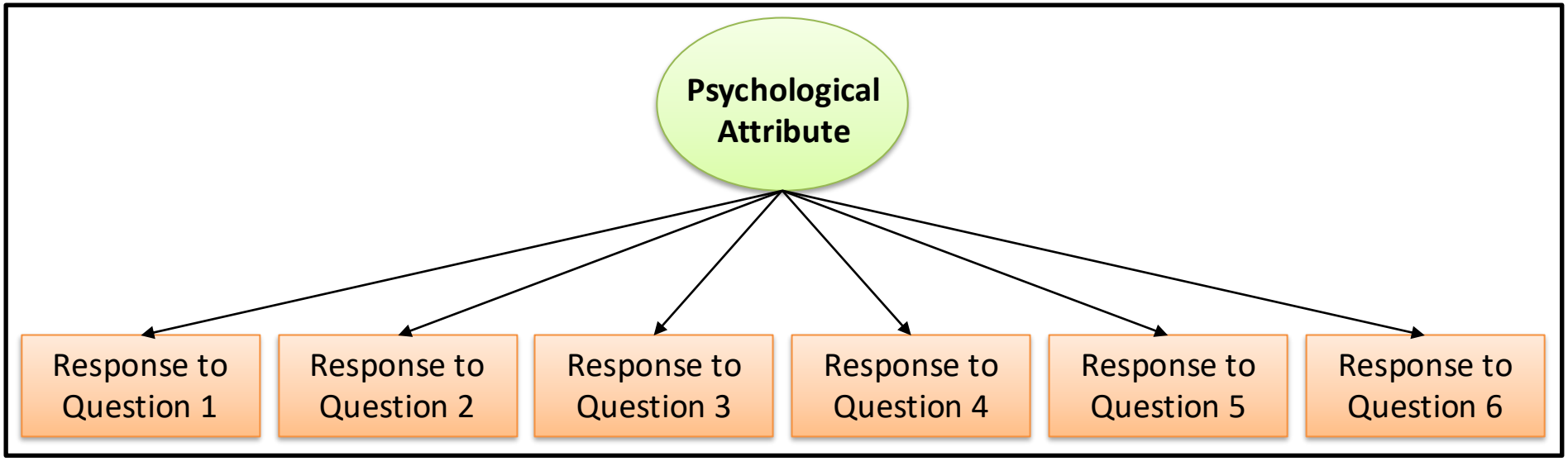
- The first question regarding test dimensionality concerns the number of dimensions reflected in a set of test items
- Some tests include items that reflect a single psychological attribute, and others include items that reflect more than one attribute
- When a psychological test includes items that reflect only a single attribute of a person, this means that responses to those items are driven only by that attribute
- In such cases, we say that the test is ***unidimensional***, because its items reflect only one psychological dimension

- Consider a multiple-choice geometry exam given in a class
- Typically, a students takes the exam and receives a score based on the number of questions that he or she answers correctly
- The student's score is then interpreted as a measure of the amount of his or her "knowledge of geometry"
- This interpretation interpretable ONLY IF:
  - All items measure geometry knowledge
  - And nothing else

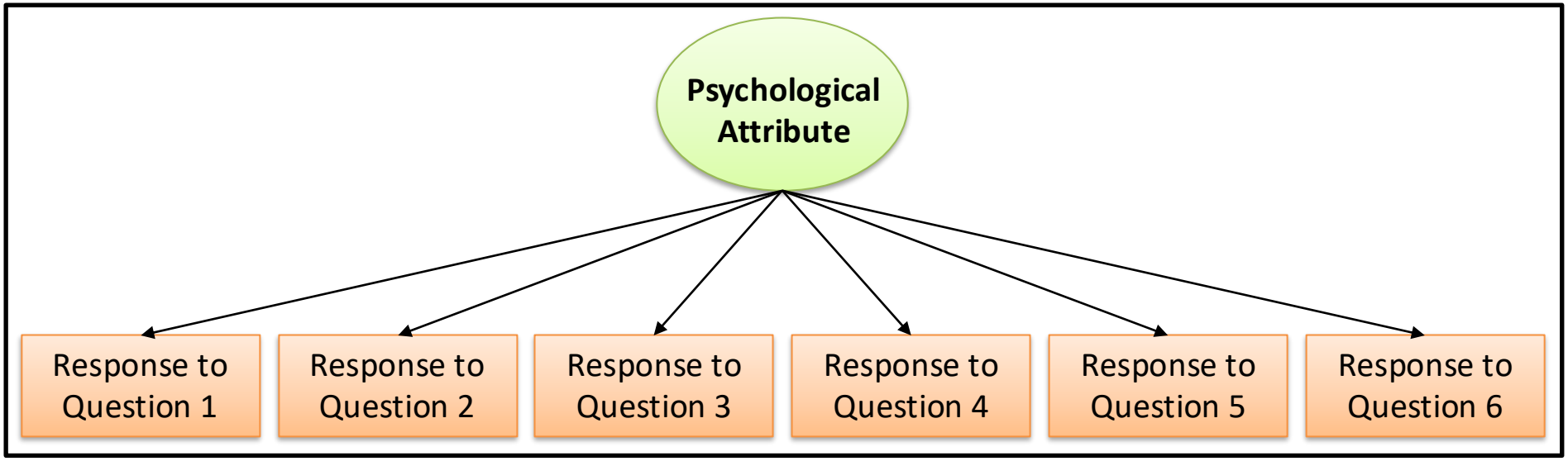
- For example, if we can believe that the test does not include algebra items, calculus items, or vocabulary items in addition to geometry items, then we can indeed have some confidence in interpreting test scores as reflecting knowledge of geometry
- That is, we could assume that the answers to each of the questions on the test are affected by that single psychological attribute
- Such a test would be thought of as unidimensional
- In addition, the test items or questions would have the property of conceptual homogeneity – responses to each item would be a function of the same psychological attribute



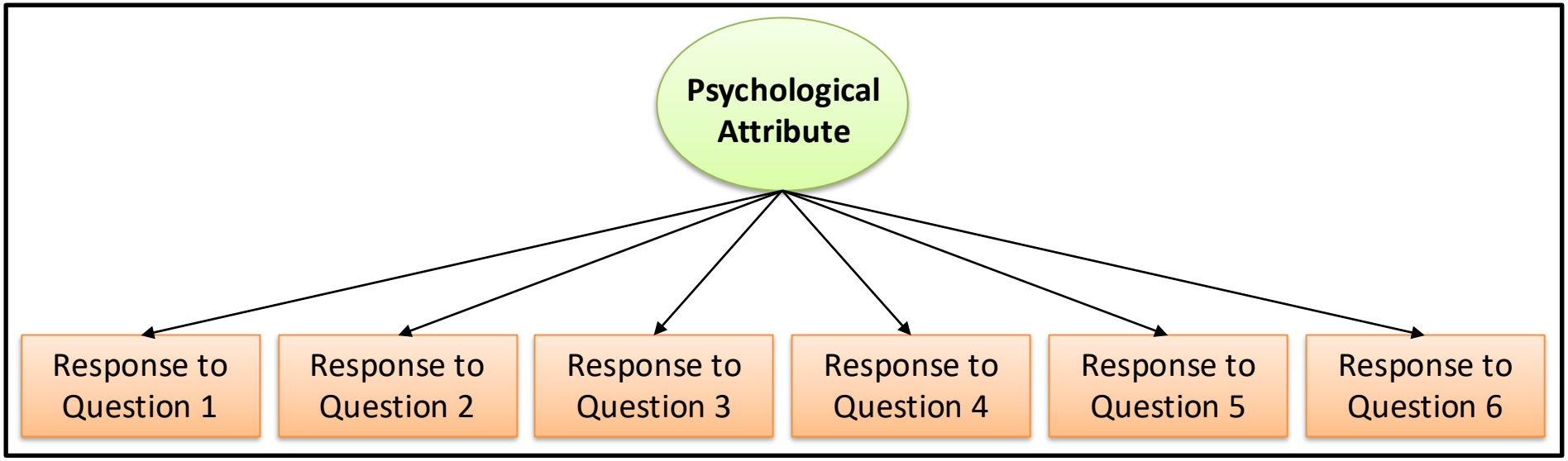
- This figure uses formatting that is standard for graphically representing a test's dimensionality
- A circle or oval represents a hypothetical psychological attribute or latent variable that affects participants' responses to test questions
- Returning to the geometry test example, the circle would represent "knowledge of geometry" because it is the psychological property that determines whether a student answers the test items correctly



- Correspondingly, squares or rectangles represent responses to each of the test questions
- Finally, the arrows' directionality represents the idea that the psychological attribute affects responses to test questions
- For example, they represent the assumption that knowledge of geometry is what affects students' answers to the test questions
- Because it shows a single psychological attribute affecting participants' responses, this figure illustrates a unidimensional test



- A test's dimensionality has implications for its scoring, evaluation, and use
- For a unidimensional test, only a single score is computed, reflecting the single psychological attribute measured by the test
- That is, all the items are combined in some way to form a composite or "total" score
- For example, if it is indeed unidimensional, the geometry test produces a single score reflecting "knowledge of geometry"



- In terms of psychometric evaluation, psychometric quality (reliability and validity) is evaluated for the single score that is obtained from a unidimensional test
- For unidimensional test, reliability and validity should be estimated and evaluated for the total score produced by the test
- In terms of test use, test users compute and interpret the total score produced by a unidimensional test

Multidimensional Tests with Correlated Dimensions

- When a psychological test includes items reflecting more than one psychological attribute, the test is considered multidimensional
- In such cases, we comfort a second dimensionality question: Are the test's dimensions associated with each other?
- The answer to this question differentiates two types of tests
- When a test has multiple dimensions that are correlated with each other, the test can be considered a ***multidimensional test with correlated dimensions***
- If dimensions are correlated, it might be because they share a more general underlying construct. This can sometimes be modeled using higher-order factors.

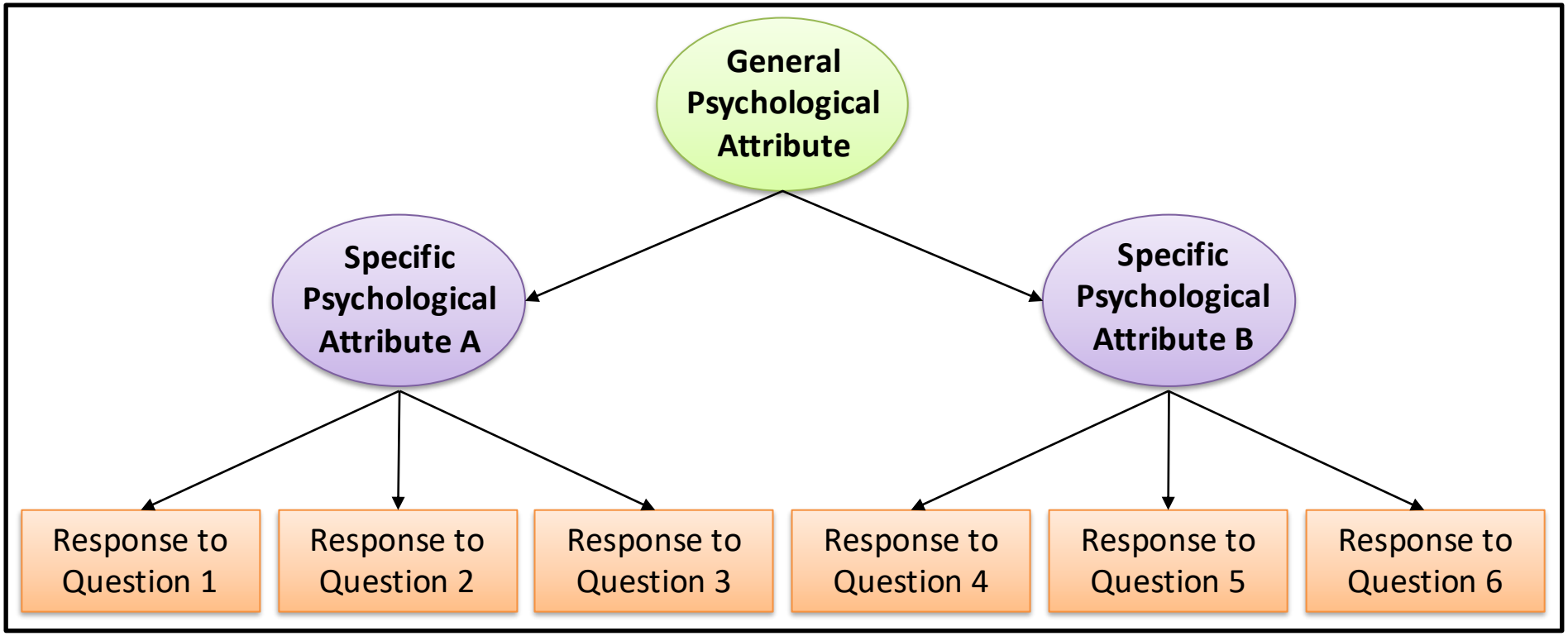
- Intelligence tests such as the Wechsler Intelligence Scale for Children (WISC-IV) and the Standford-Binet (SB5) are examples of multidimensional tests with correlated dimensions
- These tests include groups of questions that assess different psychological attributes
- The groups of questions are called ***subtests*** and they each reflect a different facet of intelligence

- For example, the SB5 has five subtests:
  - One to measure fluid reasoning
  - One to measure general knowledge
  - One to measure quantitative processing ability
  - One to measure visual-spatial processing ability
  - One that is thought to measure working memory
- Research by test developers and test evaluators has shown that the subtests of the SB5 are correlated with each other
- That is, a participant who scores relatively high on one subtest is likely to score relatively high on the other subtests as well

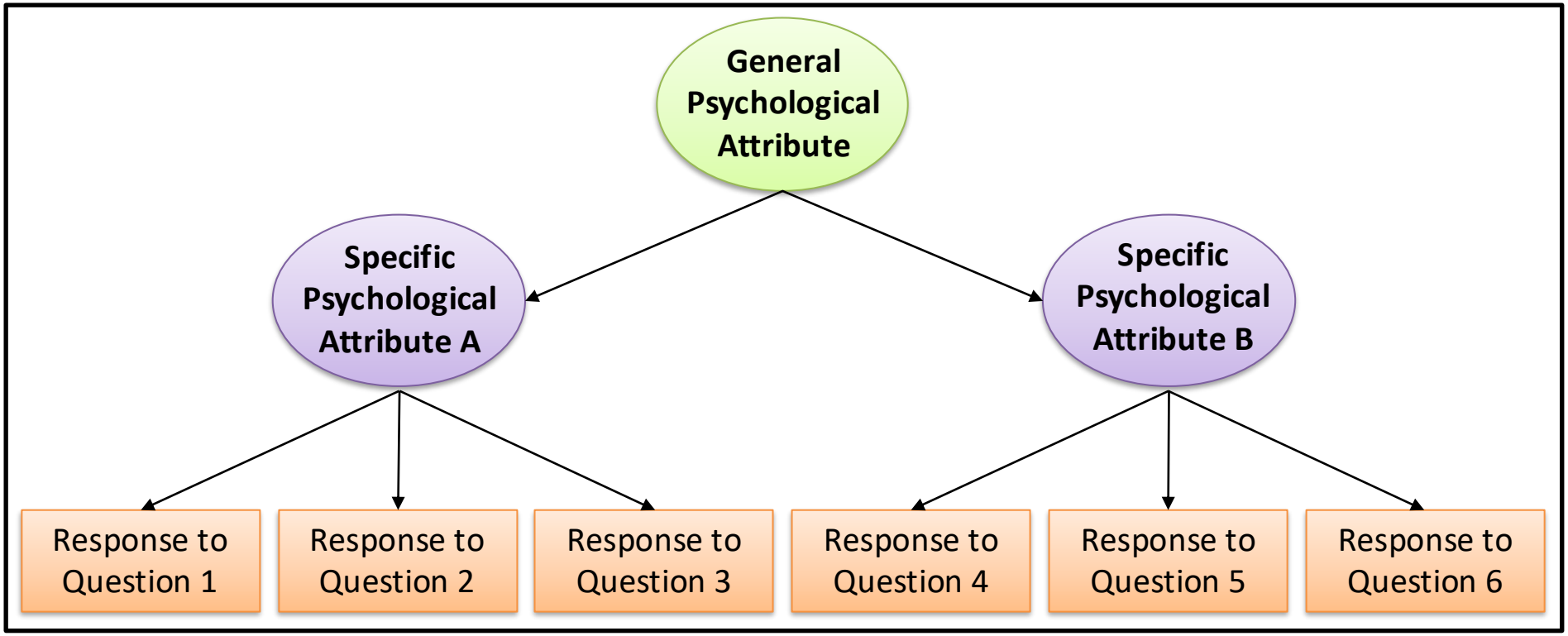
- A test’s dimensionality has important implications for the scoring, evaluating, and use of the test
- Multidimensional tests with correlated dimensions can produce a variety of scores
- Typically, each subtest has its own subtest score
- In principle, each subtest, itself, unidimensional and the questions in each subtest are conceptually homogeneous

- For example, the quantitative processing subtest of the SB5 might require a test taker to answer 10 questions
- Presumably, responses to each of those 10 questions reflect only quantitative processing and not one of the constructs represented by the other subsets
- That is, a person's responses to the 10 questions are affected only by the person's quantitative processing skills and not some other psychological attribute
- If a subtest is unidimensional, then the subset's score is interpretable with regard to a single psychological attribute

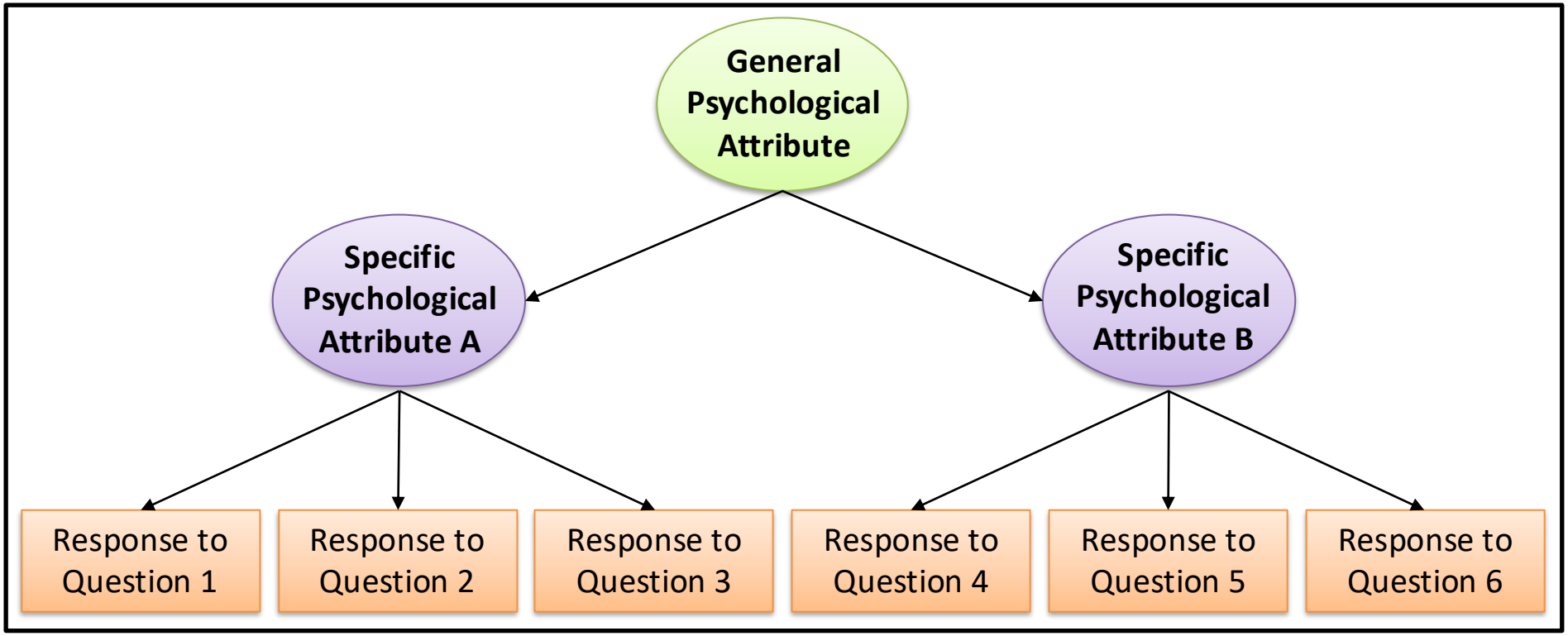
- In addition to scores for each subtest, multidimensional tests with correlated dimensions are often scored in a way that produces a total score, combined across several subtests
- That is, subtest scores are often combined with each other to produce a total test score
- For example, the five subtest scores from the SB5 are combined to form an overall “full-scale” score representing general intelligence, or g
- We can think of g as affecting a variety of more specific psychological attributes, which in turn affect the way people answer the test questions



- This type of test structure is presented in the figure
- Note that there are two levels of psychological attributes
- Responses to each question are affected by a specific attribute
- For example, an individual's responses to the questions to the quantitative processing subtest of the SB5 are affected by his or her psychological ability to process quantitative information



- In contrast, an individual's responses to the questions on the visual-spatial processing subtest of the SB5 are affected by his or her psychological ability to process visual-spatial information
- In addition to these specific psychological attributes, there is a general psychological attribute affecting each specific attribute



- For example, an individual's abilities to process quantitative information and to process visual-spatial information are partially determined by his or her general cognitive ability, or intelligence
- This general attribute is often called a ***higher-order factor*** because it is at a more general level than the specific factors or attributes

- In terms of test evaluation, multidimensional tests are different from unidimensional tests
- Recall that a unidimensional test has one and only one score and this score is evaluated with regard to its psychometric quality
- In contrast, multidimensional tests have a score for each subtest, and each subtest score is evaluated with regard to its psychometric quality
- It is possible that a multidimensional test could have some subtests that have reasonable psychometric quality and other subtests that have poor psychometric quality

- Therefore, each subtest requires psychometric examination
- For example, the developers and users of the SB5 have examined carefully the reliability and validity of each of its five subtests
- In addition, a multidimensional test with correlated dimensions may have a total test score that is computed across its subtests
- Thus, <u>this total score also requires psychometric evaluation</u>
- For example, the developers and users of the SB5 have examined the reliability and validity of its full-scale score

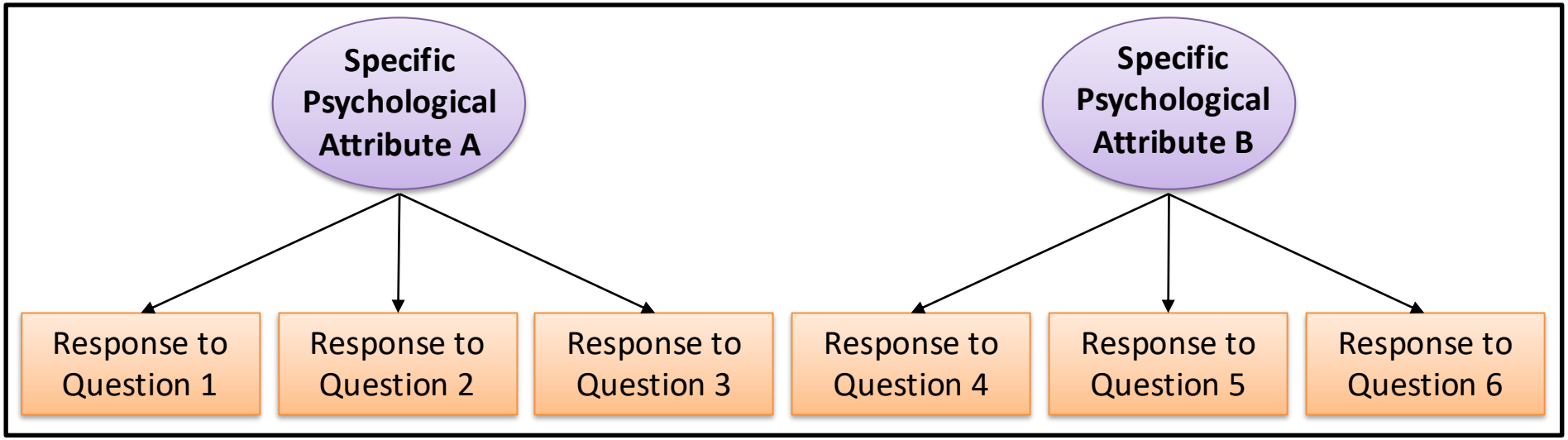
- In terms of test use, multidimensional tests offer a variety of options
- Test users could use any or all of the subtest scores, depending on their relevance to the research or practical context
- In addition, test users could use a total test score from a test with correlated dimensions if such a score is computed and has acceptable psychometric properties

Multidimensional Tests with Uncorrelated Dimensions

- The second dimensionality question regards the degree to which a multidimensional test's dimensions are associated with each other
- If a test's dimensions are not associated with each other (or are only weakly associated with each other), then the test can be considered a ***multidimensional test with uncorrelated dimensions***

- Several personality tests are multidimensional with dimensions that are generally treated as if they are uncorrelated
- For example, a test called the NEO Five Factor Inventory (NEO-FFI) is a 60-item questionnaire reflecting five dimensions of personality
- That is, the NEO-FFI is designed to measure five relatively independent personality attributes, and these five attributes are not typically treated as reflecting any higher-order factors
- Test takers receive five scores –one for each dimension- and each one is itself treated as if it were unidimensional
- In a sense, such tests could be viewed as a set of unrelated unidimensional tests that are presented with their items mixed together

- In terms of scoring, evaluation, and use, multidimensional tests with uncorrelated dimensions are similar to those with correlated dimensions, with one important exception.
- For tests with uncorrelated dimensions, no total test score is computed
- That is, a score is obtained for each dimension, but the dimensions' scores are not combined to compute a total test score
- Furthermore, each of the dimension scores is evaluated in terms of psychometric quality, and each is potentially used by researchers and practicing psychologists
- For example, the NEO-FFI produces only five scores –one for each of the five dimensions; however, no total test score is computed for the NEO-FFI



- This type of test structure is presented in the figure
- Similar to the multidimensional test with correlated dimensions, there are two psychological attribute, each one affecting responses to a set of questions
- However, in this figure, the two attributes are not linked together in any way
- This implies that the attributes are uncorrelated with each other

The Psychological Meaning of Test Dimensions

- After identifying the number of dimensions and the relationships among them, a third important issue is to understand what each dimension represents.
- Specifically, researchers must determine the psychological meaning of each dimension.
- For test scores to be interpreted accurately, evidence is needed to show what construct each dimension reflects.